

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

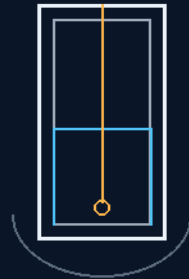
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 136

THE QUINN CENTRIFUGAL GUTTER-FEED SPORTS BOTTLE ARCHITECTURE

Spin it — the gutter gathers, the straw drinks density
the bubble lottery closed by geometry



the planet, miniaturized

DOUBLE WALL · GUTTER POCKET · FLIP-NECK

THE STRAW LIVES IN THE POCKET

Crew-hydration hardware — v1.0 in force

GP-IM-136

Revision v1.0 — 23 August 2026

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VETTED BATCH 18 — WORKS · BUBBLE TEST OWED

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 136

PHASE 136: THE QUINN CENTRIFUGAL GUTTER-FEED SPORTS BOTTLE ARCHITECTURE — STATUS: CURRENT — BUBBLE-REMOVAL PERFORMANCE TESTING OWED. The bottle that drinks like a planet. In microgravity a water bag is a lottery — the gulp you draw is as likely air as water, and every bubble swallowed is a crew comfort and digestion problem logged. Phase 136 rejects the non-rigid space-bag and the paste packet outright and engineers the container for the physics: a double-walled rigid bottle whose inner lining carries vertical gutter reduction points, so a slow spin sets the fluid climbing the walls by centrifugal force and the gutters gather it into a solid, unbroken pocket at the perimeter gate — density where the straw lives. The flip-neck sports latch anchors the straw terminal directly inside that solid-fluid pocket: un-broken density draw, every sip, no bubble lottery. It is the file's rotating-tank law shrunk to a litre — the same made-not-borrowed vector the waste bowl (137) and the shower (146) run on — dressed as consumer hardware a crew member can hold in one hand. The vet graded the liquid positioning sound; the bubble-removal performance — how clean the draw is across spin rates and fill levels — is a bench number, owed in §9.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-136, v1.0 — 23 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the bottle law as seated (contents line, the two bodies, verbatim) — §2 why the gutter works (graded) — §3 the system in the hand — §4 the doctrine record— §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 136: **Centrifugal Gutter-Feed Sports Bottle Architecture**. Rejects non-rigid plastic space-bags to eliminate microgravity hydration waste. Shapes the inner lining of a double-walled container with a downward-sloping vertical channel, utilizing a non-contact magnetic

base-stirrer to fling dense fluid outward into a perimeter straw intake port while forcing air bubbles away.'

THE MASTHEAD LINE (verbatim): 'Industrial Consumer Design: Microgravity Hydration Engineering & Non-Contact Magnetic Mixing [extracted verbatim from the combined masthead line]'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Bubble-Free Personal Fluid Delivery via Localized Centripetal Sorting [extracted verbatim from the combined masthead line]'

THE DOUBLE-WALLED CONCAVE GUTTER MATRIX (VERBATIM)

* Rationale: Rejects non-rigid plastic space-bags and paste packets to eliminate waste profiles and prevent erratic microgravity air-bubble ingestion during crew hydration cycles. * Vertical Gutter Reduction Points: Shapes the inner lining of a double-walled container with a downward-sloping channel tapering directly to a perimeter straw intake port. * Magnetic Base-Stirrer Vortices: Embeds a non-contact magnetic mixing bar inside the lower base cavity, spinning fluids to drive dense liquid outward into the gutter while forcing low-density gas bubbles into the center core space.

THE FLIP-NECK STRAW SEAL ASSEMBLY (VERBATIM)

* Intrenched Straw Terminals: Anchors the baseline drinking straw directly within the solid-fluid pocket of the perimeter gutter reduction gate to guarantee un-broken density draw. * Mechanical Flip-Neck Compression: Integrates a sports-class flip-neck silicone straw latch that compresses hard against a stop-valve when closed to secure an airtight vacuum seal. * Terrestrial Baseline Conversions: Adapts the vertical geometry into a horizontal base ring gutter for 1G applications, preventing protein sediment caking to serve as a highly viable commercial fitness product option.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

CENTRIFUGAL POSITIONING, AFFIRMED: spin the container and the fluid moves to the wall — rotating-frame physics at bottle scale, the same law the file's tanks and bowls run on. The vet's green stands: 'Centrifugal liquid positioning works.'

THE GUTTER AS COLLECTOR, AFFIRMED AS THE FILE'S OWN GEOMETRY: vertical reduction points gathering the wall-climbing sheet into a solid pocket — the rotating-tank gutter doctrine (Phase 64's bench lineage) translated into the lining of a drinks bottle.

THE STRAW-IN-THE-POCKET RULE, AFFIRMED AS THE WHOLE POINT: anchor the draw terminal inside the collected density and the bubble lottery closes by geometry, not by hope.

THE OWED NUMBER, SEATED: bubble-removal performance across spin rates and fill levels — how dry

the draw stays as the bottle empties — is a measured quantity, not an argument. Owed in §9.

§3 — THE SYSTEM IN THE HAND

- THE WALLS: double-walled rigid container — no bags, no paste packets.
- THE GUTTERS: vertical reduction points in the inner lining — the climbing sheet gathered to a pocket.
- THE POCKET: solid-fluid density at the perimeter gate — where physics puts the water.
- THE STRAW: flip-neck sports latch, terminal anchored in the pocket — un-broken density draw.
- THE SPIN: slow rotation as the whole pump — the planet, miniaturized.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE MADE-NOT-BORROWED DOCTRINE (seated across the file): the vector is manufactured by rotation — the bottle, the bowl, and the shower are one law at three scales.
- THE KILL-THE-BAG DOCTRINE (the rejection): non-rigid containers are bubble lotteries with waste profiles — consumer discipline applied to life support.
- THE GEOMETRY-NOT-HOPE DOCTRINE (the straw rule): put the terminal where the physics parks the fluid — design closes the lottery.
- THE BENCH-THE-SIP DOCTRINE (the yellow): even a bottle earns its numbers — draw dryness across the fill range is measured, not claimed.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Mold the lining: double-wall shell with gutter reduction points formed to spec.
- STEP 2 — Fit the latch: flip-neck seal and straw terminal anchored at the pocket gate.
- STEP 3 — Spin the bench: draw trials across spin rates and fill levels — bubble fraction per sip logged (§9's data).
- STEP 4 — Abuse it: drop, thermal, and cycle testing to consumer-hardware standards.
- STEP 5 — Crew the trial: usability and hydration logs across a working week.
- STEP 6 — Publish the ledger: molds, latch spec, draw-dryness curves — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 64 — the gutter bench: the rotating-tank lineage this bottle shrinks.
- PHASE 137 — the waste core (GP-IM-137): the same made vector, wet duty.
- PHASE 146 — the shower (GP-IM-146): the third scale of the same law.
- PHASE 145 — the momentum directive (GP-IM-145): what the droplet does once it leaves.
- PHASE 130 — the recirculation networks (GP-IM-130): the bubble's enemy in the lines, as here in the hand.

§8 — DO-NOT-BUILD

- DO NOT bag the water — non-rigid containers are the bubble lottery being deleted.
- DO NOT sip from the middle — the terminal lives in the pocket or the design is decoration.
- DO NOT skip the spin bench — draw dryness is owed data, not a marketing adjective.
- DO NOT proliferate the parts — one bottle, one latch, one spec: the century ship's inventory law.

§9 — OWED

OWED: the bubble-removal performance data — bubble fraction per draw across spin rates and fill levels, gutter capture efficiency included. Seats additively when benched. The 12 August naming batch is carried in the §1 header.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-136, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 23 August 2026, ninth of the 20-chapter 'ok next 20' shift (the author's order, 23 August 2026: 'ok next 20'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572 and 576): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master

record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 136